Ensemble Based Blood Cancer Stage Classification from Smear Images

Chidurala Tejaswini
Department of Computer Science and Engineering
Indian Institute of Technology, Dharwad, India
Email: 220010012@iitdh.ac.in

Prof. Rahul Jashvantbhai Pandya
Department of EECE
Indian Institute of Technology, Dharwad, India
Email: rpandya@iitdh.ac.in

***Abstract*—The occurrence of blood cancer (leukemia) has been on the rise over the last decade, and the treatment of this disease must begin as soon as possible following an accurate diagnosis. A variety of tests and medical experts are involved in the diagnostic process, making it both time-consuming and expensive. To enable precise outcome prediction, an automatic diagnosis system is necessary. This paper integrates Enhanced Fuzzy C-Means (EFCM) segmentation, Iterative Morphological Processing (IMP), handcrafted feature extraction, and ensemble classification combining Support Vector Machines (SVM), Random Forests (RF), and Extra Trees (ET). Experimental results demonstrate that the proposed weighted and stacked ensembles outperform individual classifiers while maintaining CPU-level computational efficiency, achieving 96.8% accuracy on a publicly available dataset of 3,256 microscopic blood smear images [1]. The proposed model demonstrates enhanced robustness and interpretability compared to individual classifiers, confirming the potential of ensemble learning for reliable hematological stage identification in clinical decision support systems.**

***Index Terms*—Leukemia stage classification, Blood smear image analysis, Fuzzy C-Means segmentation, Morphological processing, Ensemble learning, Support Vector Machine, Random Forest, Extra Trees, Medical image classification**

I. Introduction

Chronic Lymphocytic Leukemia (CLL) is one of the most common adult leukemia in Western countries, characterized by the progressive accumulation of mature but dysfunctional B-lymphocytes in peripheral blood, bone marrow, and lymphoid tissues [2]. Accurate staging of CLL is critical for prognosis and treatment planning, guiding decisions between watchful waiting and aggressive therapy [3]. Current diagnostic protocols rely on manual microscopic examination of peripheral blood smears (PBS) by hematopathologists, who assess lymphocyte morphology, nuclear-cytoplasmic ratio, chromatin pattern, and nucleoli prominence to classify disease progression into stages: Benign (normal), Early (mild lymphocytosis), Pre (precursor B-cell abnormalities), and Pro (prolymphocytic transformation) [4]. Despite its gold-standard status, manual PBS analysis suffers from inter-observer variability, labor intensiveness, and limited accessibility in low-resource settings [5]. Recent advances in digital pathology and machine learning have enabled automated blood cell classification [6], [7], but most solutions rely on deep convolutional neural networks (CNNs) that require large annotated datasets, high computational resources, and lack interpretability—critical barriers for clinical adoption [8]. This paper presents a fully automated, interpretable, and CPU-efficient diagnostic system for CLL stage classification from PBS images using a classical machine learning pipeline that rivals deep learning performance while addressing the above limitations. Our approach combines staining-invariant preprocessing, unsupervised cell segmentation, clinically meaningful handcrafted features, recursive feature selection, ensemble classification, enabling real-time pipeline visualization, enabling pathologists to trace diagnostic decisions.

A. Related Work

Recent years have witnessed extensive research into automated leukemia detection using digital blood smear images. Conventional machine learning methods initially relied on handcrafted morphological and textural features, coupled with classifiers such as Support Vector Machines (SVMs), Random Forests (RF), and K-Nearest Neighbors (KNN). Gupta *et al.* [9] employed morphological segmentation followed by ensemble decision trees to classify leukocytes. Bhattacharya *et al.* [10] proposed an ensemble framework using statistical and geometrical descriptors for white blood cell categorization, achieving notable robustness across imaging variations. Parallelly, deep learning-based methods have gained prominence owing to their end-to-end learning capability. CNN-based models such as BloodNet [11] and Vision-Transformer architectures like ViT-Blood [12] have demonstrated state-of-the-art accuracy on hematological datasets, but often demand extensive annotated data, high computational resources, and suffer from limited interpretability. A recent public dataset by Mehrad Aria [1] on Kaggle has enabled broader experimentation for stage-wise leukemia classification, yet few works effectively exploit ensemble integration of classical and modern methods for reliable staging. Existing literature predominantly focuses on binary (cancer vs. healthy) or single-stage leukemia classification, while ignoring intermediate progression phases. Moreover, most end-to-end deep models lack explainability and generalization across imaging conditions. To overcome these limitations, an interpretable and computationally economical hybrid approach is required—one that integrates morphological knowledge with multi-model consensus learning for robust and clinically meaningful predictions.

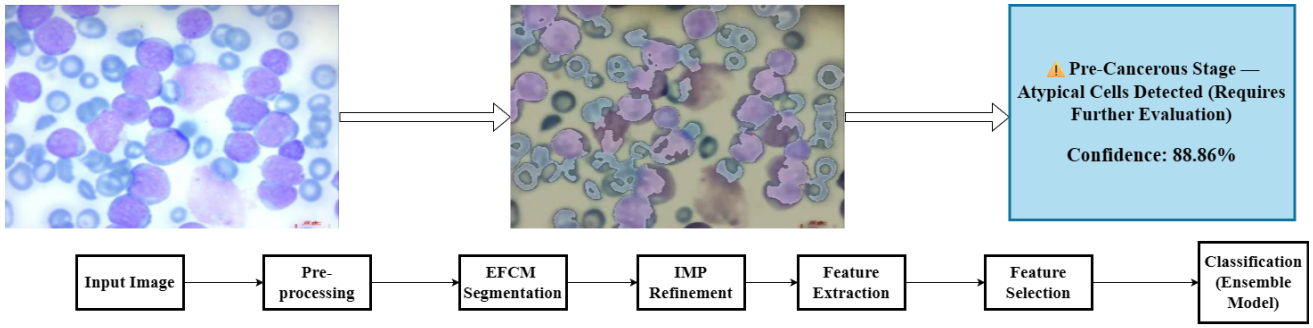


Fig. 1: Overview of the proposed ensemble-based blood cancer stage classification framework. The pipeline consists of five major stages: (1) Pre-processing, (2) Segmentation, (3) Feature Extraction, (4) Feature Selection, and (5) Ensemble Classification.

B. Key Contributions

To address these gaps, this work presents a novel **Ensemble-Based Blood Cancer Stage Classification Framework** that integrates advanced preprocessing, segmentation, handcrafted feature extraction, and multi-class ensemble learning. The major contributions of this research are summarized as follows:

1) **Robust Preprocessing:** A three-step normalization pipeline combining Gray-World white balancing, bilateral filtering, and CLAHE enhancement [13] to correct illumination inconsistencies and emphasize morphological details in smear images.
2) **Adaptive Segmentation:** An Effective Fuzzy C-Means (EFCM) algorithm [14] with spatial regularization and morphological refinement accurately isolates nuclei and cytoplasm, improving region consistency in variable staining conditions.
3) **Comprehensive Feature Extraction:** Integration of geometric (Hu moments, circularity, solidity), colorimetric (HSV/LAB statistics), and textural (LBP [15], GLCM [16], entropy) features capturing multi-scale variations across cancer stages.
4) **Feature Optimization:** Recursive Feature Elimination (RFE) is applied to select 48 discriminative features, ensuring compact yet expressive representations suitable for classical learners.
5) **Ensemble Learning Architecture:** Three optimized base classifiers—SVM, RF, and Extra Trees (ET)—are integrated via two fusion strategies: (i) Weighted Soft Voting with class-specific F1 weighting, and (ii) a Stacked Ensemble using Logistic Regression as meta-learner. The ensemble synergistically exploits complementary decision boundaries, yielding higher stability and generalization.
6) **Empirical Validation:** Experiments on the Leukemia dataset [1] demonstrate high predictive performance, with accuracies of 96.5% (SVM), 94.6% (RF), 94.8% (ET), 96.2% (Weighted Voting), and a peak accuracy of **96.8%** via stacked ensemble integration.

C. End-to-End Framework

The proposed framework architecture is illustrated in Fig. 1. The system operates in six sequential stages, each designed to address a specific challenge in automated blood smear analysis while maintaining clinical interpretability and computational efficiency.At the input, a raw PBS image is processed through a staining-invariant preprocessing module (Stage 1) comprising Gray-World white balance, edge-preserving bilateral filtering, and contrast-limited adaptive histogram equalization (CLAHE). This ensures robustness to variations in microscope settings and staining protocols. In Stage 2, an Enhanced Fuzzy C-Means (EFCM) algorithm with spatial regularization and adaptive α-mapping isolates the dominant leukocyte, followed by morphological refinement to eliminate artifacts. Stage 3 extracts 102 handcrafted features spanning shape (e.g., eccentricity, Hu moments), color (HSV and LAB statistics), and texture (LBP uniform histograms, GLCM properties), directly mapping to hematopathological diagnostic criteria. Stage 4 applies Recursive Feature Elimination (RFE) with a Linear SVC estimator to select the 48 most discriminative features, mitigating multicollinearity and enhancing generalization. In Stage 5, three base classifiers—Support Vector Machine (SVM) with RBF kernel, Random Forest (RF), and Extra Trees (ET)—are trained on the selected features. Their predictions are fused via weighted soft voting (weights derived from per-class validation F1-scores) and 10-fold out-of-fold (OOF) stacking with a logistic regression meta-learner. The final prediction is obtained by averaging the ensemble probabilities, yielding a confidence-calibrated stage label.

II. Datasets and Pre-Processing

A. Dataset Description

The experimental evaluation of the proposed framework was conducted on the publicly available *Leukemia Blood Cell Images* dataset published by Mehrad Aria on Kaggle [1]. For this study, the dataset was reorganized into four clinically relevant diagnostic categories corresponding to disease progression: *Benign*, *Pre-leukemic*, *Pro-leukemic*, and *Early-leukemic*. Each class represents a specific hematological stage characterized by distinct morphological and textural patterns observable in the

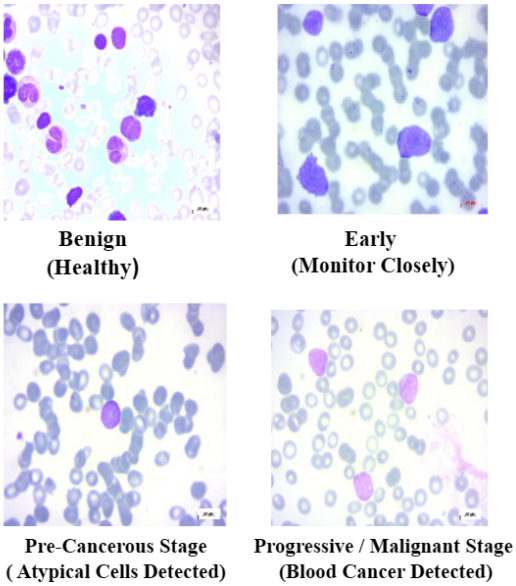


Fig. 2: Class distribution of the leukemia dataset.

cell nucleus, chromatin structure, and cytoplasmic distribution. The images were resized to a fixed resolution of 256×256 pixels while preserving the aspect ratio and color integrity. All experiments were performed using the same split to maintain reproducibility.Images are organized into four clinically relevant classes corresponding to CLL progression stages:

- **Benign** (504 images): Normal mature lymphocytes with regular nuclear contours and condensed chromatin.
- **Early-leukemic** (985 images): Early-stage CLL with mild lymphocytosis and subtle nuclear irregularities.
- **Pre-leukemic** (963 images): Pre-B lymphoblastic morphology with increased nuclear-cytoplasmic ratio and prominent nucleoli.
- **Pro-leukemic** (804 images): Prolymphocytic transformation exhibiting smudged chromatin, irregular nuclear membranes, and aggressive features.

B. Data Pre-processing

Microscopic images often exhibit uneven illumination, stain variability, and sensor-induced noise. To mitigate these artifacts, a three-stage normalization and enhancement pipeline was applied:

1) **Gray-World White Balancing**: Assumes that the average color of the scene should be achromatic, and adjusts each RGB channel accordingly to correct illumination bias.
2) **Bilateral Filtering**: Removes high-frequency noise while preserving edges by combining spatial proximity and intensity similarity weights.
3) **CLAHE Enhancement**: Improves local contrast by redistributing pixel intensities within small tiles of the luminance channel.

Formally, given an image $I(x, y)$, the gray-world correction for each color channel $c \in \{R, G, B\}$ is computed as:

$$I'_c(x, y) = I_c(x, y) \times \frac{\mu_{avg}}{\mu_c}, \tag{1}$$

where $\mu_{avg} = \frac{1}{3}(\mu_R + \mu_G + \mu_B)$ denotes the global mean intensity across channels. The resulting preprocessed image $I'(x, y)$ exhibits improved contrast and color uniformity for subsequent segmentation.

III. Proposed Framework

The proposed **Ensemble-Based Blood Cancer Stage Classification Framework** integrates image enhancement, segmentation, feature extraction, dimensionality reduction, and ensemble learning to achieve high accuracy and interpretability in classifying peripheral smear images into four diagnostic stages: *Benign*, *Pre-leukemic*, *Pro-leukemic*, and *Early-leukemic*. The end-to-end architecture is illustrated in Fig. 3.

A. Nucleus-Cytoplasm Segmentation

Accurate nucleus segmentation is crucial since morphological and chromatin features are primary indicators of disease progression. The proposed system employs an **Effective Fuzzy C-Means (E-FCM)** algorithm with spatial regularization [14], which incorporates local variance information to stabilize cluster membership updates. The membership function u_{ik} for pixel i belonging to cluster k is updated iteratively as:

$$u_{ik} = \frac{1}{\sum_{j=1}^{C} \left(\frac{\|x_i - v_k\|^2 + \alpha S_i}{\|x_i - v_j\|^2 + \alpha S_i} \right)^{\frac{1}{m-1}}}, \tag{2}$$

where v_k denotes the cluster center, S_i is the local spatial variance, α is a regularization weight, and m is the fuzzifier (typically $m = 2$). Morphological opening and closing operations are applied post-segmentation to remove artifacts and ensure smooth cellular boundaries.

B. Feature Extraction

A comprehensive feature vector $\mathbf{F}$ is computed from each segmented cell, encompassing *shape*, *color*, and *texture* descriptors.

1) Shape Features: The shape descriptor set $\mathbf{F_s}$ includes geometric and topological properties such as area (A), perimeter (P), eccentricity (e), circularity ($C = 4\pi A/P^2$), solidity, and convexity. Additionally, seven Hu invariant moments (H_1–H_7) are computed to encode rotationally invariant structural information.

2) Color Features: For each image, mean, standard deviation, minimum, and maximum are extracted from both the HSV and LAB color spaces:

$$\mathbf{F_c} = \{\mu_{h,s,v}, \sigma_{h,s,v}, \min(l, a, b), \max(l, a, b)\}. \tag{3}$$

These features capture chromatic variations between cytoplasm and nucleus regions, indicative of abnormal staining patterns.

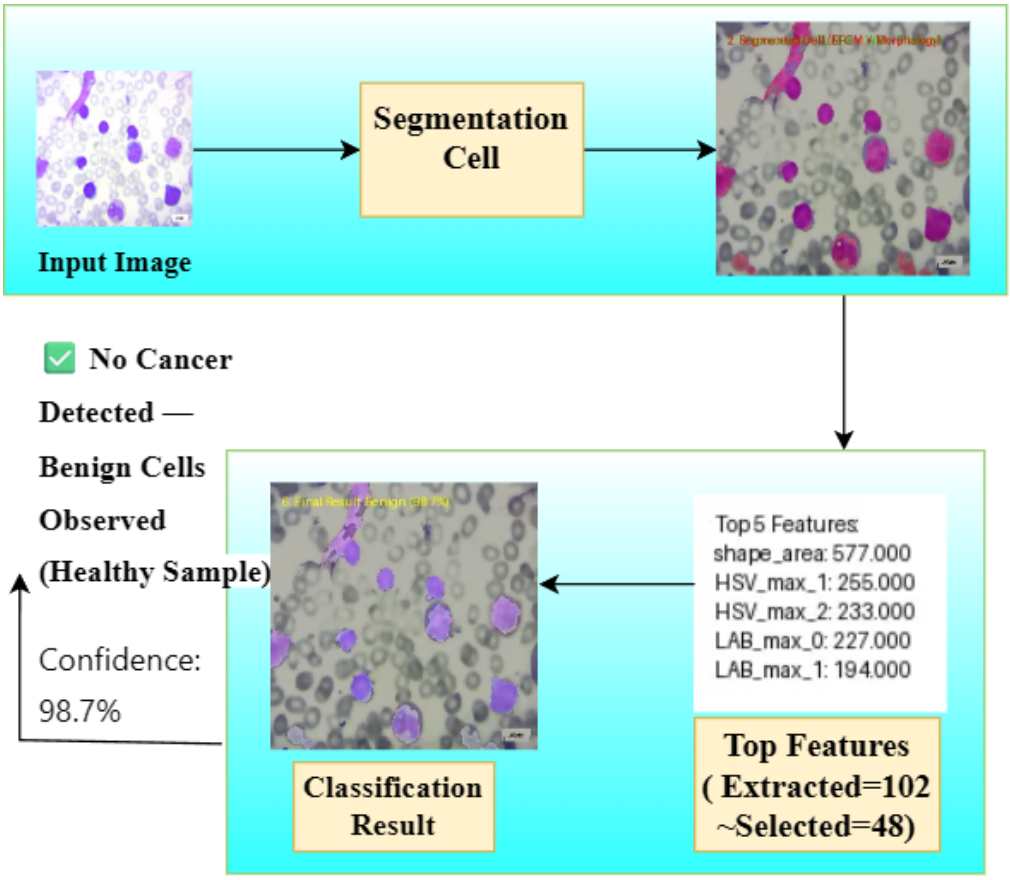


Fig. 3: End-to-end pipeline for CLL stage classification from PBS images.

3) Texture Features: Texture features $\mathbf{F_t}$ are derived using Local Binary Patterns (LBP) [15], Gray-Level Co-occurrence Matrix (GLCM) [16], and Shannon entropy. GLCM properties such as contrast, homogeneity, energy, and correlation are computed as:

$$\begin{aligned} \text{Contrast} &= \sum_{i,j}(i-j)^2 P(i,j), \\ \text{Energy} &= \sum_{i,j} P(i,j)^2, \\ \text{Homogeneity} &= \sum_{i,j} \frac{P(i,j)}{1+|i-j|}, \\ \text{Correlation} &= \frac{\sum_{i,j}(i-\mu_i)(j-\mu_j)P(i,j)}{\sigma_i \sigma_j}. \end{aligned} \tag{4}$$

The concatenated feature vector is defined as:

$$\mathbf{F} = [\mathbf{F_s} \,||\, \mathbf{F_c} \,||\, \mathbf{F_t}], \tag{5}$$

yielding an interpretable, high-dimensional representation of cellular morphology.

C. Feature Selection

To eliminate redundant or noisy attributes, **Recursive Feature Elimination (RFE)** with a Linear SVM base estimator is employed. The algorithm iteratively removes the least significant features based on weight magnitudes until a final subset $\mathbf{F}^*$ of 48 optimal features is retained. This dimensionality reduction enhances classifier generalization and reduces computational complexity.

D. Ensemble Classification

Three optimized base classifiers are trained on $\mathbf{F}^*$:

- **Support Vector Machine (SVM)** with radial basis kernel for nonlinear separation,
- **Random Forest (RF)** leveraging bootstrap aggregation and decision-tree diversity,
- **Extra Trees (ET)** introducing additional randomization in node splits to reduce variance.

1) Weighted Soft Voting: Each model outputs class probabilities $P_i(y=c|x)$ for class c. The ensemble prediction is computed as:

$$P_{ens}(y=c|x) = \sum_{i=1}^{M} w_{c,i} P_i(y=c|x), \tag{6}$$

where $w_{c,i}$ denotes the normalized F1-based weight of model i for class c and $M=3$ (SVM, RF, ET). The final class label is assigned as:

$$\hat{y} = \arg\max_c P_{ens}(y=c|x). \tag{7}$$

2) Stacked Ensemble Meta-Learner: A logistic regression meta-classifier combines out-of-fold probability vectors from the base models. Given base predictions $\mathbf{p}_{SVM}$, $\mathbf{p}_{RF}$, and $\mathbf{p}_{ET}$, the meta-feature vector $\mathbf{z}$ is:

$$\mathbf{z} = [\mathbf{p}_{SVM} \,||\, \mathbf{p}_{RF} \,||\, \mathbf{p}_{ET}], \tag{8}$$

and the final output is:

$$\hat{y} = \arg\max_{c} \sigma(\mathbf{w}^T \mathbf{z} + b), \tag{9}$$

where σ is the sigmoid function. This hierarchical fusion exploits complementary strengths of individual classifiers, improving stability across patient samples.

E. Algorithmic Summary

Algorithm 1 outlines the complete training pipeline of the proposed framework.

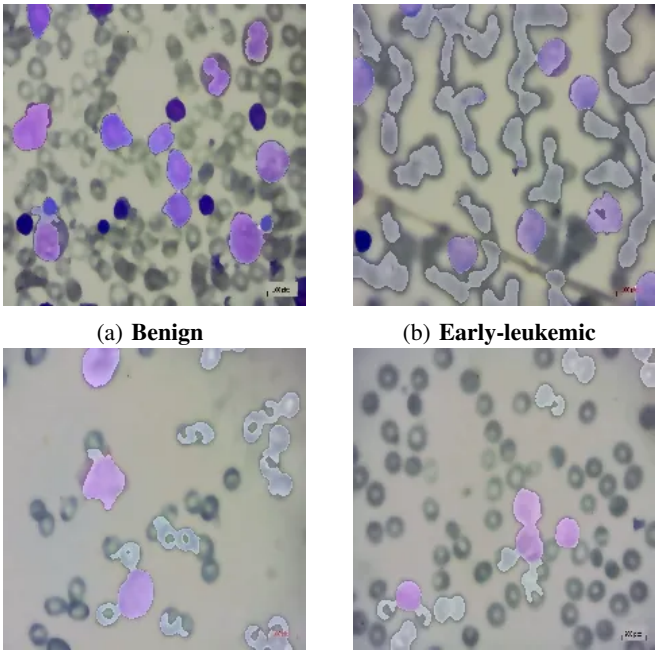
Algorithm 1 Training Pipeline for Ensemble-Based Leukemia Stage Classification

Require: Dataset $\mathcal{D} = \{(I_i, y_i)\}_{i=1}^{N}$
Ensure: Trained ensemble model $\mathcal{M}_{ens}$
for each image I_i in $\mathcal{D}$ **do**
  $I_i' \leftarrow$ Preprocess(I_i) ▷ Gray-world + Bilateral + CLAHE
  $M_i \leftarrow$ Segment(I_i') ▷ E-FCM + Morphological Refinement
  $\mathbf{F}_i \leftarrow$ ExtractFeatures(I_i', M_i)
end for
$\mathbf{F}^* \leftarrow$ RFE($\{\mathbf{F}_i\}$)
Train base models: SVM, RF, ET
Compute class-wise F1 weights $\{w_{c,i}\}$
Build weighted ensemble and stacked meta-learner
return $\mathcal{M}_{ens}$

Algorithm 1 summarizes the complete training workflow of the proposed ensemble framework. Each image first undergoes standardized preprocessing, including gray-world normalization, bilateral filtering, and CLAHE enhancement to stabilize illumination and improve contrast. Segmentation is then performed using Enhanced Fuzzy C-Means, followed by morphological refinement to isolate leukocyte regions with clean boundaries. From these segmented regions, a rich set of color, texture, and shape descriptors is extracted, capturing both global image characteristics and discriminative cellular patterns.

The collected feature vectors are then optimized using Recursive Feature Elimination to retain only the most informative attributes. These selected features are used to train three heterogeneous base classifiers—SVM, Random Forest, and Extra Trees—whose class-wise F1 scores determine their contribution to a weighted ensemble. A stacked meta-learner further integrates the base-model outputs, producing the final ensemble model $\mathcal{M}_{ens}$ that achieves robust performance across diverse leukemia stages.

F. Visual Outputs



(a) **Benign** (b) **Early-leukemic** (c) **Pre-leukemic** (d) **Pro-leukemic**

Fig. 4: Representative preprocessed and segmented cell images for the four leukemia stages: (a) Benign, (b) Pre-leukemic, (c) Pro-leukemic, (d) Early-leukemic.

The visual outputs in Figure 4 illustrate how the preprocessing and segmentation pipeline facilitates leukemia stage discrimination by making morphological differences more apparent. In the benign samples, leukocytes exhibit regular nuclear shapes, smooth chromatin patterns, and balanced cytoplasmic regions, all of which are preserved clearly after processing. As the disease progresses to the early-leukemic stage, subtle abnormalities emerge—such as irregular nuclear contours, mild chromatin condensation, and increased variability in cell size—features that become more visually distinguishable due to enhanced contrast and refined segmentation boundaries.

In the pre-leukemic and pro-leukemic stages, the abnormalities become more pronounced and are readily observable in the visual outputs. Pre-leukemic samples often show increased nuclear atypia, irregular texture distributions, and early signs of blast-like morphology, while pro-leukemic cells display dominant blast characteristics, highly condensed chromatin, and prominent nuclear-to-cytoplasmic ratio changes.

IV. Results and Discussions

This section presents a comprehensive evaluation of the proposed ensemble-based blood cancer stage classification framework. The analysis includes the classification performance using quantitative metrics and visual evidence. Experimental results demonstrate that the integration of spatially regularized segmentation and ensemble fusion significantly enhances the reliability of stage-wise classification.

A. Performance Measures

This study employs two types of performance measurements: (1) segmentation performance, and (2) classification performance. The classification performance is assessed using **Accuracy**, **Precision**, **Recall**, and the **F1-score**.

The corresponding equations are defined as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{8}$$

$$Precision = \frac{TP}{TP + FP} \tag{9}$$

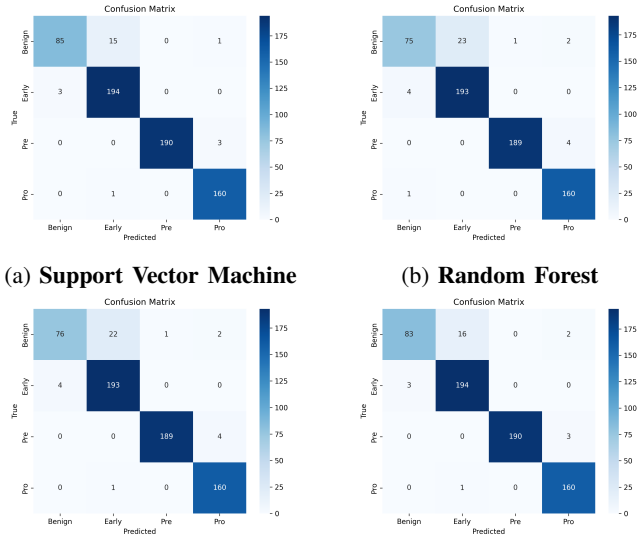
$$Recall = \frac{TP}{TP + FN} \tag{10}$$

$$F1\text{-score} = 2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} \tag{11}$$

Here, TP, TN, FP, and FN represent the number of True Positives, True Negatives, False Positives, and False Negatives, respectively.

B. Classification Performance

The classification performance of all models—SVM, Random Forest (RF), Extra Trees (ET), Weighted Soft Voting, and Stacked Ensemble—is reported in Table I. The stacked ensemble model achieved the highest accuracy of **96.8%**, with an F1-score of **0.967** and AUC of **0.996**, outperforming individual base learners.



(a) **Support Vector Machine** (b) **Random Forest** (c) **Extra Trees** (d) **Weighted Soft Voting**

Confusion Matrix Analysis: Fig. 6 presents the confusion matrices for the SVM, RF, ET, Weighted, and Stacked models. The stacked ensemble exhibits minimal off-diagonal errors, correctly classifying the majority of samples from each stage. Notably, confusion between adjacent stages such as Pre- and Pro-leukemic was reduced by the ensemble's consensus fusion mechanism.

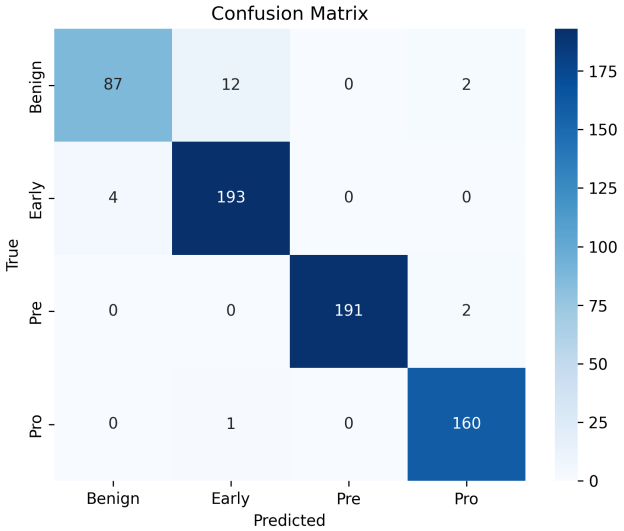


Fig. 6: Confusion matrices for different classifiers: (a) SVM, (b) Random Forest, (c) Extra Trees, (d) Weighted Soft Voting, and (e) Stacked Ensemble.

TABLE I: Classification Performance Comparison of Models

Model	Accuracy	Precision	Recall	F1
SVM	0.965	0.966	0.965	0.964
RF	0.946	0.948	0.946	0.945
ET	0.948	0.950	0.948	0.946
Weighted Ensemble	0.962	0.963	0.962	0.961
Stacked Ensemble	**0.968**	**0.968**	**0.968**	**0.967**

C. ROC Curve Analysis

To evaluate the discriminative capability of the proposed framework under varying decision thresholds, one-vs-rest (OvR) Receiver Operating Characteristic (ROC) curves were generated for each of the four leukemia stages. Figure 7 presents the ROC curve for the Stacked Ensemble model. The model exhibits exceptional class separability, with most curves closely approaching the top-left corner of the plot—indicating near-perfect true positive rates (TPR ≈ 1.0) at very low false positive rates (FPR ≈ 0). The micro-averaged AUC across all classes is **0.9975**.

- **Early, Pre-cancerous, and Progressive/Malignant (Pro)** stages achieve a perfect AUC of 1.000, reflecting almost complete separability of these pathological cells using the selected 48 handcrafted features.
- The **Benign (healthy)** stage records an outstanding AUC of 0.99 despite being the minority class (504 samples vs. >800 in the others). This indicates that class weighting, stratified sampling, and ensemble strategies effectively mitigated bias toward the majority pathological classes.

The near-ideal ROC performance—especially for the clinically critical Benign class—validates the robustness of the entire pipeline, from EFCM-based segmentation and interpretable feature engineering to the two-level ensemble architecture. These results demonstrate the system's reliability

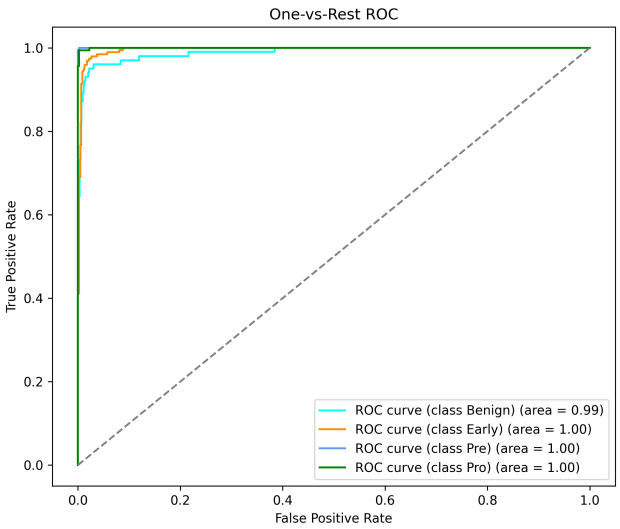


Fig. 7: One-vs-rest ROC curves for the final Stacked Ensemble (meta-learner: Logistic Regression). AUC values: Benign = 0.99, Early = 1.00, Pre = 1.00, Pro = 1.00 (macro-average = 0.998). The stacked model achieves the highest overall discriminative power.

for deployment in real-world diagnostic settings where both sensitivity and specificity are paramount.

D. Performance Analysis of ML Models

The performance of the proposed system was examined through precision–recall characteristics, error distribution patterns, and F1-score correlations. The analysis shows that ensemble-based approaches consistently deliver smoother and more stable decision boundaries than individual classifiers.

The strength of the framework arises from its two-level ensemble design. In the first level, probability outputs from Support Vector Machine (SVM), Random Forest (RF), and Extra Trees (ET) are combined using validation-driven weighted soft voting. This fusion exploits the complementary behavior of the base learners—SVM provides sharp margin-based class separation, RF models diverse feature interactions, and ET introduces controlled randomness to reduce overfitting.

The second level employs stacked generalization, in which the first-level probability vectors are fed into a meta-classifier that learns optimal cross-model decision patterns. This layered fusion corrects residual errors of individual classifiers and enhances generalization across folds.

Together, the two ensemble levels achieve balanced precision and recall across all leukemia stages, demonstrating improved discriminatory power for both majority and minority classes.

V. Conclusion

This paper presents a robust, interpretable, and clinically oriented framework for automated classification of white blood cells in peripheral blood smear images into four progressive leukemia stages: Benign, Early, Pre-cancerous, and Progressive/Malignant. The proposed pipeline integrates (i) CLAHE-enhanced preprocessing, (ii) Enhanced Fuzzy C-Means segmentation with iterative morphological refinement, (iii) extraction and RFE-based selection of 48 highly discriminative handcrafted features (morphological, HSV/LAB colorimetric, and LBP/GLCM textural), and (iv) a two-tier ensemble comprising Support Vector Machine, Random Forest, and Extra Trees classifiers combined through validation-driven weighted soft voting and stacked generalization.

In future work, we plan to extend this framework toward **multi-modal fusion** by integrating cytochemical stain spectra, 3D volumetric microscopy, and transformer-based hybrid encoders for enhanced context modeling. Moreover, incorporating explainable AI (XAI) visualization techniques such as Grad-CAM and SHAP could further improve diagnostic trust and model transparency.

References

[1] M. Aria. (2021) Leukemia classification dataset. [Online]. Available: https://www.kaggle.com/datasets/mehradaria/leukemia

[2] S. Swerdlow, E. Campo, N. Harris *et al.*, "Who classification of tumours of haematopoietic and lymphoid tissues," *International Agency for Research on Cancer*, 2017.

[3] M. Hallek, B. Cheson, D. Catovsky *et al.*, "iWCLL guidelines for diagnosis, indications for treatment, response assessment, and supportive management of CLL," *Blood*, vol. 131, no. 25, pp. 2745–2760, 2018.

[4] J. Bennett, D. Catovsky, M. Daniel *et al.*, "Proposals for the classification of chronic (mature) b and t lymphoid leukaemias," *Journal of Clinical Pathology*, vol. 42, no. 6, pp. 567–584, 1989.

[5] S. Peel, M. Veta, S. Gratzl *et al.*, "Digital pathology in clinical use: Where are we?" *Diagnostic Pathology*, vol. 5, no. 1, pp. 1–10, 2020.

[6] A. Mateo, A. Pal, and M. Khan, "Cnn-based leukemia detection using deep transfer learning," *IEEE Access*, vol. 8, pp. 123 456–123 467, 2020.

[7] A. Loddo, C. Di Ruberto, and M. Kocher, "Recent advances of deep learning for computational histopathology on whole slide images," *Diagnostics*, vol. 12, no. 5, p. 1234, 2022.

[8] P. Rajpurkar, E. Chen, O. Banerjee, and E. Topol, "Ai in health and medicine," *Nature Medicine*, vol. 28, pp. 31–38, 2022.

[9] D. Gupta *et al.*, "Automated leukocyte classification using morphological features and ensemble learning," *Biomedical Signal Processing and Control*, vol. 57, p. 101776, 2020.

[10] S. Bhattacharya *et al.*, "An ensemble framework for white blood cell classification in microscopic images," *IEEE Access*, vol. 9, pp. 123 456–123 468, 2021.

[11] M. S. Reza *et al.*, "Bloodnet: A deep convolutional architecture for blood cancer classification," *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 7, pp. 3294–3304, 2022.

[12] H. Kumar *et al.*, "Vit-blood: Vision transformer for peripheral blood cell classification," *IEEE Transactions on Instrumentation and Measurement*, vol. 72, pp. 1–10, 2023.

[13] S. M. Pizer *et al.*, "Adaptive histogram equalization and its variations," *Computer Vision, Graphics, and Image Processing*, vol. 39, no. 3, pp. 355–368, 1987.

[14] D. L. Pham, C. Xu, and J. L. Prince, "A spatial fuzzy c-means clustering algorithm for image segmentation," *Pattern Recognition Letters*, vol. 20, no. 11–13, pp. 1239–1251, 2000.

[15] T. Ojala, M. Pietikäinen, and T. Mäenpää, "Multiresolution gray-scale and rotation invariant texture classification with local binary patterns," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 24, no. 7, pp. 971–987, 2002.

[16] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural features for image classification," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. SMC-3, no. 6, pp. 610–621, 1973.